



DWF_dp_simd_mult functions

SIMD multiply

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Description

Revision History

The DWF_dp_simd_mult functions implement a configurable SIMD multiplier. They allow you to either multiply arguments *a* and *b* as full-width vectors (for example, one 32-bit multiplication) or to multiply smaller partitions of *a* and *b* using multiple parallel multipliers (for example, two 16-bit multiplications or four 8-bit multiplications). The argument *no_confs* specifies the number of possible configurations, and argument *conf* dynamically selects one of the configurations. Configuration with number *conf* has 2^{conf} partitions of size $\text{width}/2^{\text{conf}}$. Arguments *a* and *b* and the return value are all either unsigned or signed (two's complement).

Table 1-1 Function Names

Function Name	Description
DWF_dp_simd_mult	VHDL unsigned SIMD multiply
DWF_dp_simd_mult	VHDL signed (two's complement) SIMD multiply
DWF_dp_simd_mult_uns	Verilog unsigned SIMD multiply
DWF_dp_simd_mult_tc	Verilog signed (two's complement) SIMD multiply

Table 1-2 Argument Description

Argument Name	Type	Width / Values	Description
<i>a</i>	Vector ^a	width	Input multiplier
<i>b</i>	Vector ^a	width	Input multiplicand
<i>no_confs</i>	Integer	≥ 2	Number of configurations (VHDL only, constant)
<i>conf</i>	Vector ^b	$\text{ceil}(\log_2[\text{no_confs}])$	Configuration selection: 2^{conf} partitions of size $\text{width}/2^{\text{conf}}$
DWF_dp_simd_mult	Vector ^a	width	Return value

a. unsigned or signed in VHDL

b. std_logic_vector in VHDL

Table 1-3 Parameter Description (Verilog)

Parameter	Values	Description
<i>width</i>	≥ 2 , must be a multiple of $2^{\text{no_confs}-1}$	Word length
<i>no_confs</i>	≥ 2	Number of configurations

Verilog Include File: DW_dp_simd_mult_function.inc

Functional Description

```
z[2*width-1:0] = DWF_dp_simd_mult (a[width-1:0], b[width-1:0], no_confs,
                                   conf[bit_width(no_confs)-1:0])

conf = 0:
  z[2*width-1:0]          = a[width-1:0]          * b[width-1:0]
conf = 1:
  z[2*width-1:2*width/2] = a[width-1:width/2]    * b[width-1:width/2]
  z[2*width/2-1:0]       = a[width/2-1:0]         * b[width/2-1:0]
conf = 2:
  z[2*width-1:2*width*3/4] = a[width-1:width*3/4] * b[width-1:width*3/4]
  z[2*width*3/4-1:2*width/2] = a[width*3/4-1:width/2] * b[width*3/4-1:width/2]
  z[2*width/2-1:2*width/4] = a[width/2-1:width/4] * b[width/2-1:width/4]
  z[2*width/4-1:0]         = a[width/4-1:0]       * b[width/4-1:0]
...
```

Example: width = 32, no_confs = 3

```
conf = 0:
  z[63: 0] = a[31: 0] * b[31: 0]
conf = 1:
  z[63:32] = a[31:16] * b[31:16]
  z[31: 0] = a[15: 0] * b[15: 0]
conf = 2:
  z[63:48] = a[31:24] * b[31:24]
  z[47:32] = a[23:16] * b[23:16]
  z[31:16] = a[15: 8] * b[15: 8]
  z[15: 0] = a[ 7: 0] * b[ 7: 0]
```

For more information about the DesignWare datapath functions, see to [DesignWare Datapath Functions Overview](#).

Related Topics

- [DesignWare Datapath Functions Overview](#)
- [DesignWare Building Block IP User Guide](#)

VHDL Example

```

library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use IEEE.numeric_std.all;
use DWARE.DW_dp_functions.all;
-- DWARE.DW_dp_functions_arith package if IEEE.std_logic_arith is used

entity DWF_dp_simd_mult_test is
  port (op1, op2  : in  signed(31 downto 0);
        config_no : in  std_logic_vector(1 downto 0);
        product   : out signed(63 downto 0));
end DWF_dp_simd_mult_test;

architecture rtl of DWF_dp_simd_mult_test is
begin
  product <= DWF_dp_simd_mult (a => op1, b => op2,
                              no_confs => 3, conf => config_no);
end rtl;

```

Verilog Example

```

module DWF_dp_simd_mult_test (op1, op2, config_no, product);

  input  signed [31:0] op1, op2;
  input          [1:0] config_no;
  output signed [63:0] product;

  // Passes the parameters to the function
  parameter width      = 32;
  parameter no_confs = 3;

  // add "$SYNOPSISYS/dw/sim_ver" to the search path for simulation
  `include "DW_dp_simd_mult_function.inc"

  assign product = DWF_dp_simd_mult_tc (op1, op2, config_no);

endmodule

```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
March 2021	DWBB_202009.4	■ Corrected operator symbols in Example on page 2 ■ Added this Revision History table and the document links on this page

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